

Scalability & Future Plan

Date	03 July 2026
Team ID	-
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	AI features depend on an active Google Gemini API and internet connection.	AI services become unavailable if the API or internet is unavailable.	High
2	Large PDF/DOCX files may increase processing time.	Longer response time while extracting and summarizing document content.	Medium
3	The application is currently optimized for web browsers only.	Mobile users have a reduced experience compared to a dedicated mobile application.	Low

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	AI Request Handling	Direct synchronous calls to Gemini per request	Add request queueing, response caching, and asynchronous processing for improved performance.
2	Database	Single MongoDB Atlas cluster	Enable database indexing, sharding, and backup strategies as user data grows.
3	Deployment	Local development environment / Single server deployment.	Deploy using Docker containers with cloud hosting and load balancing for scalability.
4	Authentication	JWT with fixed expiry	Add OAuth (Google Login), Multi-Factor Authentication (MFA), and refresh token support.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Mobile-responsive UI improvements and performance optimization.	5 days	Improved usability and faster response times.
Phase 2	Voice-based AI Tutor with speech recognition and text-to-speech.	5 days	Interactive hands-free learning experience.
Phase 3	AI Flashcards, Spaced Repetition, and Personalized Study Recommendations.	5 days	Better knowledge retention and personalized learning.
Phase 4	Mobile Application (Android/iOS), Classroom Collaboration, and Offline Learning Support.	5 days	Increased accessibility, collaboration, and broader user adoption.